Understanding 3D Pyramid Vertices

Using Simple PyOpenGL Coordinates

IIT Patna

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What is a Vertex?

A vertex is simply one corner point of a 3D object.

$$\boxed{\text{Vertex} = (x,\ y,\ z)}$$

- x tells left or right.
- y tells up or down.
- z tells front or back.

$$\boxed{+x = \text{right}, \quad -x = \text{left}}$$

$$\boxed{+y = \text{up}, \quad -y = \text{down}}$$

$$\boxed{+z = \text{front}, \quad -z = \text{back}}$$

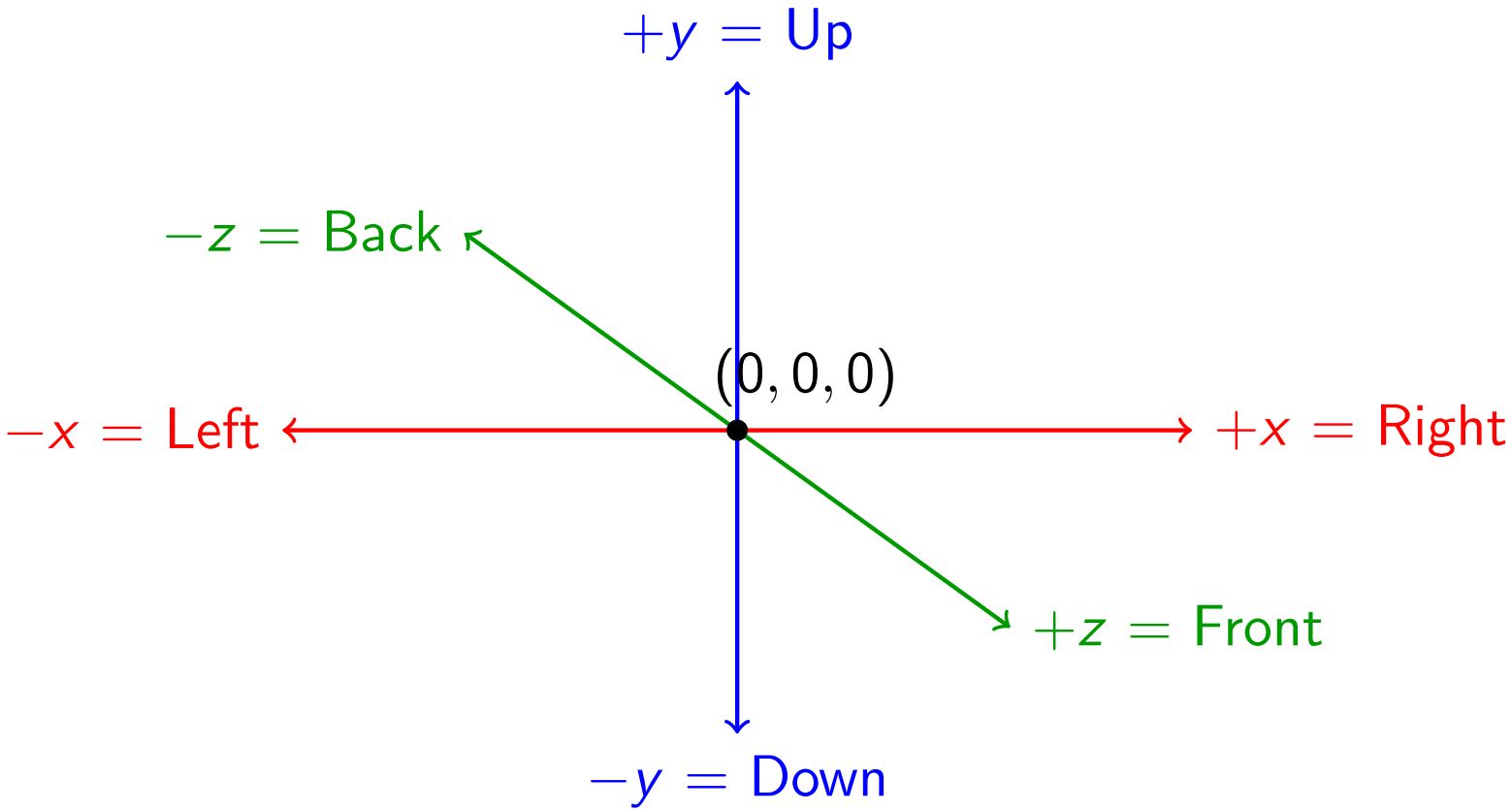
Pyramid Vertices in Python

```
vertices = [
    [1, -1, -1],    # 0
    [1, -1,  1],    # 1
    [-1, -1, 1],    # 2
    [-1, -1, -1],   # 3
    [0,  1,  0]     # 4 top
]
```

Simple Meaning

The first four vertices make the bottom square base.
The fifth vertex is the top point of the pyramid.

Very Easy Sign Rule



$$\boxed{x = \text{Left/Right}, \quad y = \text{Up/Down}, \quad z = \text{Front/Back}}$$

Step 1: Construct the Bottom Base

A pyramid first needs a square base.

$$\boxed{y = -1}$$

Why?

Very Simple Reason

All four points of the base are at the same bottom level.
Therefore, all four vertices use the same value:

$$y = -1$$

The only values that change are x and z.

Vertex V_0

$$\boxed{V_0 = (1, -1, -1)}$$

Read the coordinate one value at a time:

Coordinate	Value	Meaning
x	$+1$	Right
y	-1	Bottom
z	-1	Back

Therefore,

$$\boxed{V_0 = \text{Right} + \text{Bottom} + \text{Back}}$$

Vertex V_1

$$\boxed{V_1 = (1, -1, 1)}$$

Coordinate	Value	Meaning
x	$+1$	Right
y	-1	Bottom
z	$+1$	Front

Therefore,

$$\boxed{V_1 = \text{Right} + \text{Bottom} + \text{Front}}$$

Vertex V_2

$$\boxed{V_2 = (-1, -1, 1)}$$

Coordinate	Value	Meaning
x	-1	Left
y	-1	Bottom
z	$+1$	Front

Therefore,

$$\boxed{V_2 = \text{Left} + \text{Bottom} + \text{Front}}$$

Vertex V_3

$$\boxed{V_3 = (-1, -1, -1)}$$

Coordinate	Value	Meaning
x	-1	Left
y	-1	Bottom
z	-1	Back

Therefore,

$$\boxed{V_3 = \text{Left} + \text{Bottom} + \text{Back}}$$

The Four Base Vertices

Vertex	Coordinate	Meaning
V_0	$(1, -1, -1)$	Right, Bottom, Back
V_1	$(1, -1, 1)$	Right, Bottom, Front
V_2	$(-1, -1, 1)$	Left, Bottom, Front
V_3	$(-1, -1, -1)$	Left, Bottom, Back

Very Important Observation

All four have:

$$\boxed{y = -1}$$

because all four belong to the bottom base.

Base Viewed From the Top

BACK: $z = -1$



FRONT: $z = +1$

Vertex V_4: The Top Point

The final vertex is:

$$\boxed{V_4 = (0, 1, 0)}$$

Coordinate	Value	Meaning
x	0	Center left/right
y	$+1$	Up / Top
z	0	Center front/back

Therefore,

$$\boxed{V_4 = \text{Top Center}}$$

Why is the Top Vertex $(0, 1, 0)$?

Imagine the center of the square base:

$$\boxed{(0, -1, 0)}$$

To create the top of the pyramid, we keep:

$$x = 0$$

and

$$z = 0$$

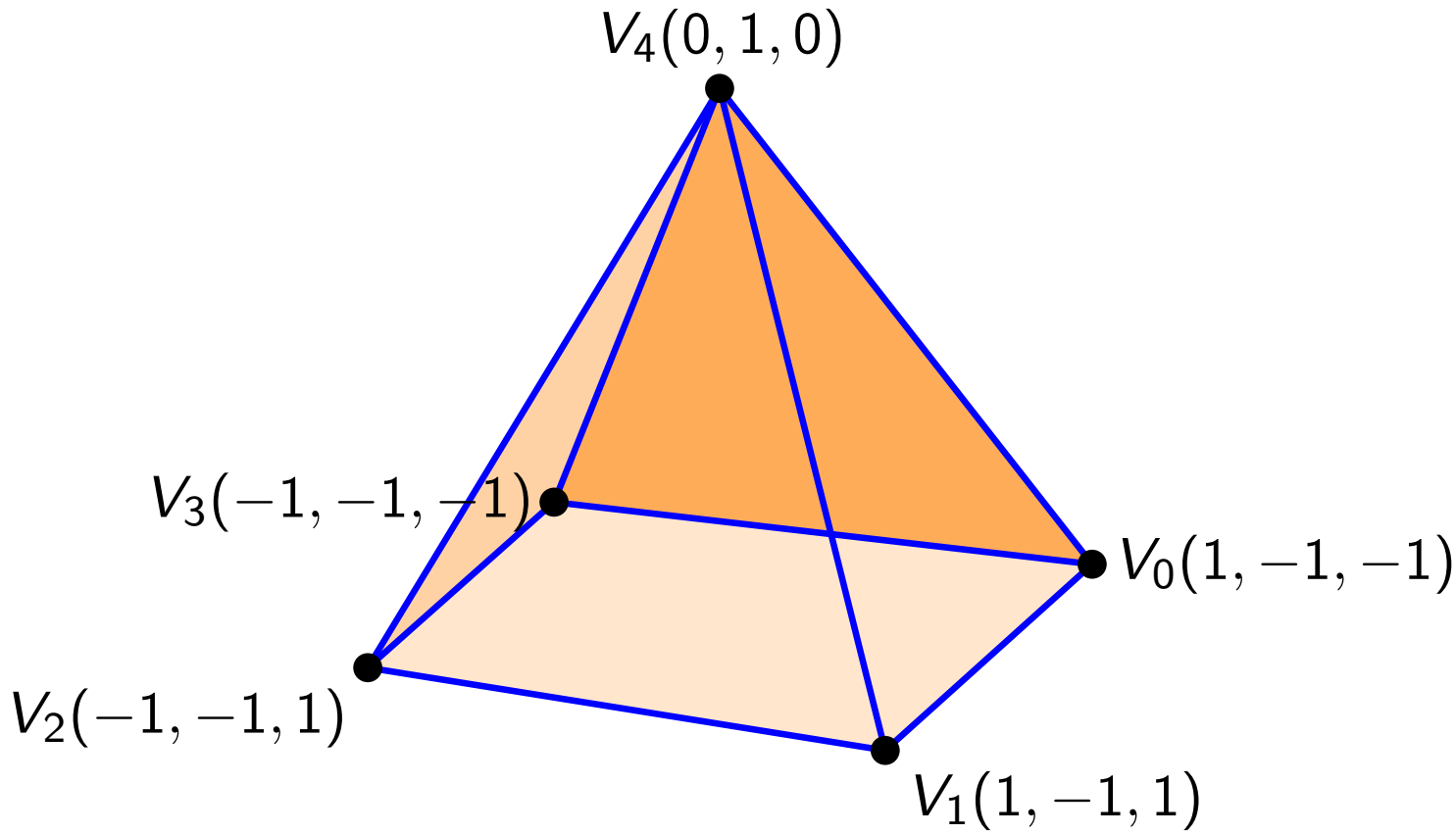
because the top should stay in the center.
Then we move only upward:

$$y = 1$$

Therefore,

$$\boxed{V_4 = (0, 1, 0)}$$

Complete 3D Pyramid with Coordinates



How Were These Coordinates Created?

We do not need to memorize them.
Start with the base:

$$\boxed{(\pm 1, -1, \pm 1)}$$

This means:

- $x = \pm 1$ gives left or right.
- $y = -1$ keeps all points at the bottom.
- $z = \pm 1$ gives front or back.

Therefore the four combinations are:

$$(1, -1, -1)$$

$$(1, -1, 1)$$

$$(-1, -1, 1)$$

The Easiest Memory Trick

$$\boxed{\begin{array}{rcc} +x & = & \text{Right} \\ -x & = & \text{Left} \\ +y & = & \text{Up} \\ -y & = & \text{Down} \\ +z & = & \text{Front} \\ -z & = & \text{Back} \end{array}}$$

For example:

$$(-1, -1, 1)$$

means:

$$\boxed{\text{Left} + \text{Bottom} + \text{Front}}$$

So this is vertex V_2.

Final Pyramid Vertices

$$
\begin{aligned}
V_0 &= (1, -1, -1) \\
V_1 &= (1, -1, 1) \\
V_2 &= (-1, -1, 1) \\
V_3 &= (-1, -1, -1) \\
V_4 &= (0, 1, 0)
\end{aligned}
$$

Main Idea

Four points create the square bottom.
One point at the center and above creates the top.
Connecting them creates the 3D pyramid.

Thank You